A Global CIT Model Fusing Ground-Based GNSS and Space-Borne LEO Satellite Data for Monitoring the Geomagnetic Storm

Tianyang Hu, Xiaohua Xu, and Jia Luo

***Abstract*— The global computerized ionospheric tomography (CIT) based on the global navigation satellite system (GNSS) is an effective tool for monitoring the ionospheric responses to space weather events. However, the main challenge of global CIT is the ill-posed problem due to the unevenly distributed GNSS stations. In this study, we propose a global CIT model fusing ground-based GNSS and space-borne low Earth orbit (LEO) satellite data. The slant total electron content (STEC) observations from more than 600 GNSS stations and 34 LEO satellites are jointly used as the data source for CIT. The simulation experiment shows that compared with CIT based on only GNSS data (CIT_G), CIT based on GNSS + LEO data (CIT_{GL}) improves the voxel penetration rate by 18.31%. In the real data experiment, the root mean square error (RMSE) of the STEC obtained by CIT_{GL} is reduced by 24.59% on average compared to that obtained by CIT_G. The ionospheric electron density (IED) profiles and the total electron content (TEC) maps derived from CIT_{GL} results are of better accuracies than those derived from CIT_G results. CIT_{GL} is further applied for investigating the ionospheric variations around the geomagnetic storm on day of year (DOY) 308, 2021, and the results demonstrate that CIT_{GL} effectively reconstructs the 3-D distributions and the evolutions of the global ionospheric disturbances around the storm. In particular, CIT_{GL} captures the uplift of the hmF2 during the main phase of the storm.**

***Index Terms*— Computerized ionospheric tomography (CIT), geomagnetic storm, global navigation satellite system (GNSS), low Earth orbit (LEO) satellites.**

I. Introduction

IONOSPHERIC electron density (IED) is an important parameter to describe ionospheric conditions. The precise monitoring of IED contributes to understanding the Earth's upper atmosphere and the solar-terrestrial environment. The computerized ionospheric tomography (CIT) is an important tool for the 3-D imaging of IED at a low computational cost. Austen et al. [1] first proposed the CIT technique using total electron content (TEC) data collected at multiple receivers for obtaining the IED image. After that, many theoretical and experimental studies were carried out [2], [3], [4], [5]. With the development of the global navigation satellite system (GNSS), GNSS-based CIT has been successfully applied in reconstructing the 3-D IED distributions during different space weather events, such as geomagnetic storms [6], [7], [8] and earthquakes [9], [10]. Currently, most related studies focus on the regional CIT models. However, since geomagnetic storms usually cause ionospheric disturbances over the globe, it is valuable to establish a high-precision CIT model on a global scale for comprehensively monitoring the variations of IED. The main challenge of the global CIT model is the ill-posed problem. In the conventional CIT model using ground-based GNSS, due to the uneven distributions of GNSS stations, especially very few stations in the ocean area, a large number of voxels are not traversed by GNSS rays and cannot be corrected, which results in the large condition number of the normal equation in the CIT algorithm. Despite the considerable efforts in the past decades to improve the CIT iterative algorithms, the ill-posed problem cannot be effectively solved in the regions severely lacking observations [11], [12].

Multisource data fusion is an effective method to solve the problem by improving the number of observations. With the increasing number of low Earth orbit (LEO) satellites, it is expected to provide valuable supplementary space-borne observations for the ground-based GNSS. Most of the LEO satellites are equipped with the GNSS radio occultation (RO) receivers, which are capable of receiving GNSS signals without limitations of time and geographical location, thus filling the data gap of the ground-based GNSS over the ocean area. Besides, the near-horizontal rays from GNSS satellites to LEO satellites are not limited by the elevation angles, which means that the LEO data has much higher vertical resolution than the near-vertical ground-based GNSS rays. In recent years, the LEO data from some RO missions have been used as the source data or constraints in the CIT experiments [13], [14], [15], [16], [17], and the results show that the incorporation of LEO data can improve the estimation accuracy of the peak heights and the vertical resolutions of the CIT results. However, most of the above studies only introduced data from single RO mission into the CIT, with a much smaller data volume compared with the ground-based GNSS. Moreover, most of the RO missions operated in the past two decades consist

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of only a few LEO satellites, which provide limited amount of data, and their contribution to the accuracy improvements of the CIT results are also limited.

In the last few years, some new RO missions consisting of more LEO satellites or providing much larger data volumes have been put in operation. The Constellation Observing System for the Meteorology, Ionosphere, and Climate-2 (COSMIC-2) is a follow-on RO mission of COSMIC, with six LEO satellites launched in June 2019, providing around 9000 slant TEC (STEC) observations per day, which are about four times those of COSMIC. In addition, the Spire Global has more than 120 LEO CubeSats, more than 40 of which have been collecting RO measurements since November 2019 [18], providing around 2000 STEC observations per day till now. These large numbers of space-borne ionospheric observations should help to further improve the voxel penetration rate and the geometry of GNSS rays in the CIT reconstruction. Ren et al. [19] evaluated the performance of LEO-augmented CIT using the simulated observations from the designed LEO constellation, showing the higher accuracy of the GNSS + LEO solution than the GNSS-only solution. Mei et al. [20] combined the real data from 460 GNSS stations and 11 LEO satellites to reconstruct the global IED distribution. Cheng et al. [21] conducted the CIT experiment using multi-GNSS and COSMIC measurements to observe the global ionospheric disturbances during the geomagnetic storm. Nevertheless, currently, there is no case of combining the observations from ground-based GNSS and the recent new RO missions such as COSMIC-2 and Spire satellites to construct a global CIT model.

In this work, we propose a global CIT model fusing ground-based GNSS and space-borne LEO satellite data. The STEC data from the new RO missions with higher data volumes is employed in the CIT model, and the vertical TEC (VTEC) data are combined with STEC data to further improve the voxel penetration rate. The feasibility and superiority of the proposed model are assessed in both the simulation experiment and real data experiment, and the accuracy of CIT results are validated with the reference STEC data and COSMIC-2 IED profiles. Then, the model is applied to monitor the 3-D distribution of the ionospheric disturbances during a geomagnetic storm occurring in 2021.

II. Methodology

A. CIT Theory

In the voxel-based CIT strategy, the inversed region is divided into lots of small voxels, and the IED of each voxel is assumed as a constant in the time period of CIT. The ith STEC can be expressed as the sum of IED in the voxels traversed by the ith GNSS ray

$$\mathrm{STEC}_i = \sum_{j=1}^{n} A_{ij} x_j + \varepsilon_i \tag{1}$$

where x_j is the IED of the jth voxel, A_{ij} is the intercept of the ith GNSS ray path through the jth voxel, n is the number of all voxels in the CIT region, and ε_i is the noise of the ith STEC. The STEC observed in the CIT time period can be written in matrix form

$$\underset{m\times 1}{\mathrm{STEC}} = \underset{m\times n}{A} \cdot \underset{n\times 1}{x} + \underset{m\times 1}{\varepsilon} \tag{2}$$

where m is the number of STEC, A is the design matrix, x is a column vector of the IED in each voxel, and ε is the noise vector.

Due to the ill-posed problem, (2) is commonly solved by the iterative algorithms rather than being solved directly. Here, we select the simultaneous iteration reconstruction technique (SIRT) to estimate the IED of each voxel. SIRT corrects the IED of voxels after considering all the ray paths, rather than one ray path at a time, which avoids the overcorrection of the voxels traversed by multiple rays [22]. The IED of the jth voxel is corrected by SIRT as follows:

$$x_j^{k+1} = x_j^k + \lambda \cdot \frac{1}{R_j} \sum_{i=1}^{R_j} \frac{A_{ij}\left(\mathrm{STEC}_i - \sum_{j=1}^{n_i} A_{ij} x_j^k\right)}{\sum_{j=1}^{n_i} A_{ij}^2} \tag{3}$$

where x_j^{k+1} and x_j^k are the IED of the jth voxel after the $(k+1)$th and kth iteration step, respectively, R_j is the number of rays traversing the jth voxel, n_i is the number of voxels traversed by the ith ray, λ is the relaxation factor which is typically in the range of $0 < \lambda < 1$, and it is set as 0.2 in this work. The iteration is repeated until the maximum difference between the IED values before and after the last iteration is less than 0.03×10^{11} el/m^3 [23].

B. Global CIT Model Combining VTEC Data

The CIT area in this study ranges from 90°S to 90°N in latitude, from 180°W to 180°E in longitude, and from 100 to 5000 km in altitude. We divide the area into voxels with the horizontal resolution of 2.5° × 5°, while with varying vertical resolutions at different altitudes. Specifically, due to the significant variation of IED around the F2 peak height (hmF2), the resolution is set as 10 km at altitudes from 200 to 450 km, and 50 km at other altitudes. It implies that there are 118 layers in total, each with 5184 voxels. Considering the smaller data volume of RO missions than the ground-based GNSS, the time windows of the ground-based and space-borne STEC are set as 3 and 5 min before and after the experimental moments, respectively. Compared with Mei et al. [20], our model utilizes a smaller time window, while is of higher spatial resolution, which is due to that larger volume of RO data from multiple missions employed in this study enables more voxels to be traversed by GNSS rays even during a shorter time period.

To further improve the data volume, the VTEC data are used besides the STEC. VTEC is derived from STEC by the single layer mapping function

$$\begin{cases} \mathrm{VTEC} = \mathrm{MF} \cdot \mathrm{STEC} \\ \mathrm{MF} = \cos\left(\arcsin\left(\dfrac{R}{R + H_{\mathrm{shell}}}\right)\sin(\alpha z)\right) \end{cases} \tag{4}$$

where R is the Earth radius, z is the satellite zenith angle at the receiver, $H_{\mathrm{shell}} = 506.7$ km is the single layer shell height, and $\alpha = 0.9782$. Similar to STEC, VTEC is also

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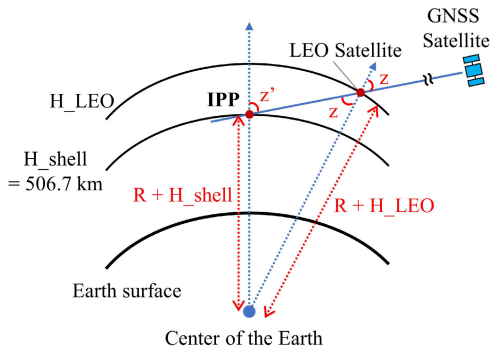


Fig. 1. Schematic of the computation of the satellite zenith z' angle at the IPP.

applied to (2), while it is used to correct a column of voxels rather than a slant set of voxels. As verified by Prol et al. [24] and Zheng et al. [25], the utilization of VTEC data increases the number of observations, especially in the regions lacking STEC observations, while does not significantly affect the overall accuracy of CIT. Actually, in the following experiments, the number of VTEC observations reaches 70.98% of that of STEC observations, while the residuals of VTEC are comparable to those of STEC.

Another issue is to calculate the coordinate of the ionospheric pierce point (IPP) of VTEC. For the ground-based GNSS receivers and LEO satellites with orbital altitudes lower than H_{shell}, IPP is the intersection between the ray path and H_{shell}. While for the LEO satellites with orbital altitudes higher than H_{shell}, as shown in Fig. 1, IPP is the intersection between the ray extension and H_{shell}, and the zenith angle z' at the IPP can be computed from z

$$z' = \arcsin\left(\frac{R + H_{\text{LEO}}}{R + H_{\text{shell}}} \times \sin z\right) \tag{5}$$

where H_{LEO} is the orbital altitude of the LEO satellite. VTEC of this part of LEO satellites is derived using (4) by replacing z with z'. The inspection of datasets reveals that the average horizontal distance between IPPs and LEO satellites is about 41 km, which is significantly smaller than the voxel resolution of 2.5°. It can be considered that in our method, the locations of IPPs agree well with the locations of the column of voxels that should be corrected.

III. Datasets

A. Ground-Based GNSS Data

As shown in Fig. 2, more than 600 GNSS stations from the International GNSS Service (IGS) are selected to provide STEC observations with 30-s intervals. STEC is estimated from the carrier phase smoothed pseudoranges of GNSS dual-frequency observations as follows:

$$\text{STEC} = \frac{f_1^2 f_2^2}{40.3\left(f_1^2 - f_2^2\right)}\left(\tilde{p}_2 - \tilde{p}_1 + B^T + B^R\right) \tag{6}$$

where f_1 and f_2 are the frequencies of the GNSS signals, $\tilde{p}_1$ and $\tilde{p}_2$ are the carrier phase smoothed pseudoranges, B^T and B^R are the differential code biases (DCBs) of the satellites and receivers, which are estimated according to Jin et al. [26].

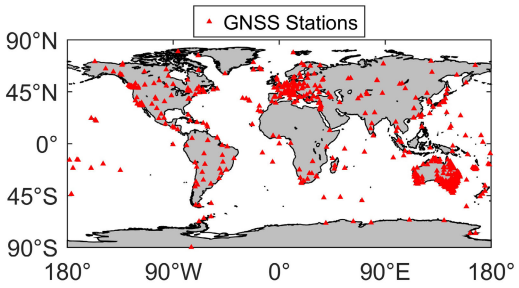


Fig. 2. Distributions of IGS GNSS stations (red triangles).

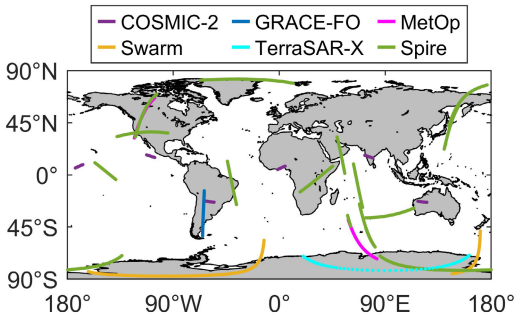


Fig. 3. Subsatellite points of LEO satellites in the time window of 10 min centered at 18:00 UT on DOY 308, 2021.

To minimize the multipath effects, the cutoff elevation angle of ground-based GNSS STEC is set as 30°.

B. Space-Borne LEO Satellite Data

The space-borne RO missions employed in this study include COSMIC-2, Gravity Recovery and Climate Experiment Follow-On (GRACE-FO), MetOp, Swarm, TerraSAR-X, and Spire. All the STEC observations with 1-s interval are along the rays from GNSS satellites to LEO satellites. Due to the temporary unevenly distributed configuration of the Spire constellation [27], only 20 Spire satellites provide STEC data in 2021, which means that a total of 34 LEO satellites are involved in our experiment. Fig. 3 shows the subsatellite points of LEO satellites in the time window of 10 min centered at 18:00 UT on day of year (DOY) 308, 2021. It is illustrated that the RO data partially covers the gaps of ground-based GNSS data over the ocean area, and Spire provides more data than other RO missions due to its larger number of LEO satellites. To ensure the confidence of LEO STEC observations, a quality control process is taken to remove the STECs with negative value or without attached DCBs estimated by the data archive centers of the RO missions.

IV. Results and Discussion

A. Simulation Experiment

It is necessary to conduct the simulation experiment to verify the superiority of the CIT model based on GNSS + LEO data (CIT_{GL}) over that based on only GNSS data (CIT_{G}) before applying the model to the real data. The simulation experiments are performed at 6:00, 12:00, and 18:00 UT on DOY 308, 2021, and the design matrix A is calculated based on the coordinates of satellites and receivers in each time

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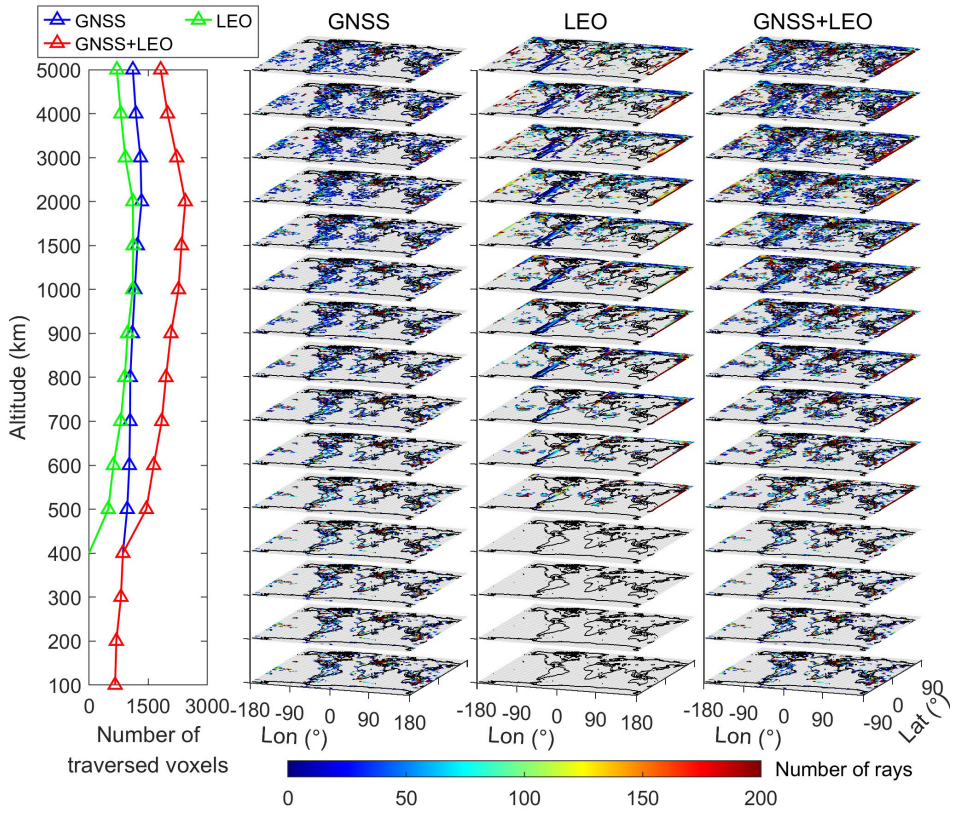


Fig. 4. Number of voxels traversed by rays and the number of rays in the traversed voxels at 6:00 UT on DOY 308, 2021, for GNSS, LEO, and GNSS + LEO data, respectively.

TABLE I

NUMBER OF THE TRAVERSED VOXELS AND THE VOXEL PENETRATION RATE FOR CIT_{GL} AND CIT_G AT 6:00 UT ON DOY 308, 2021

	CIT_{GL}	CIT_G	Increase
Number of the Traversed Voxels	116,121	228,089	111,968
Voxel Penetration Rate	18.98%	37.29%	18.31%

window. The IED provided by the IRI-plas model [28] is used as the true value, and STEC is simulated by integrating the IED true value along the GNSS ray path and adding 10% random noise. The IED background is set by adding 0.8 times the true value and 20% random noise.

Fig. 4 shows the number of voxels traversed by rays and the number of rays in the traversed voxels at 6:00 UT, for GNSS, LEO, and GNSS + LEO data, respectively, and Table I gives the number of the traversed voxels and the voxel penetration rate for CIT_{GL} and CIT_G at this moment. It is shown in Fig. 4 that LEO data fill in some voxels in the ocean and polar area that are not traversed by ground-based GNSS rays. Although the number of traversed voxels for LEO data is less than that for GNSS data, compared with using GNSS data only, fusing GNSS and LEO data improves the voxel penetration rate from 18.98% to 37.29%, with the increase reaching 18.31%. This improvement is higher than that obtained by Mei et al. [20], which is 12.54%, even though the number of voxels in the present study is nearly seven times larger. This should be attributed to the new RO missions employed and the approach of combining VTEC data adopted by us. Note that there is no voxel traversed by space-borne GNSS rays below 400 km, due to that the orbital altitudes of the LEO satellites employed are all higher than 400 km.

To examine the accuracy of CIT_{GL} and CIT_G, 5% of the ground-based and 5% of the space-borne STEC are randomly selected as the reference STEC, respectively, since the locations of LEO satellites vary with time. The reference STEC is not involved in the simulation experiment and is compared with the corresponding STEC derived from CIT_{GL} and CIT_G. The root mean square error (RMSE) is calculated as

$$\mathrm{RMSE} = \sqrt{\frac{1}{N}\sum_{i=1}^{N}\left(\mathrm{STEC}_i^{\mathrm{CIT}} - \mathrm{STEC}_i^{\mathrm{ref}}\right)^2} \tag{7}$$

where N is the number of STEC observations, and $\mathrm{STEC}_i^{\mathrm{CIT}}$ and $\mathrm{STEC}_i^{\mathrm{ref}}$ are the ith CIT-derived STEC and reference STEC, respectively. Fig. 5 shows the variations of the RMSE of CIT_{GL} and CIT_G with CIT iteration times at the three moments. Due to the higher data volume and voxel penetration rate after incorporating LEO data, the RMSEs of CIT_{GL} are generally much lower than those of CIT_G at all the three moments. Besides, CIT_{GL} has a higher convergence rate than CIT_G during the iteration process. After the first iteration, the RMSE of CIT_{GL} decreased by 0.40/0.42/0.38 TECU at the three moments, respectively, which are larger than 0.36/0.39/0.36 TECU of the RMSE of CIT_G. Fig. 5 demonstrates that fusing GNSS and LEO data significantly improves both the accuracy and the efficiency of CIT.

Fig. 6 further illustrates the 3-D maps of the errors of CIT_{GL} and CIT_G results relative to the IED true values at the three

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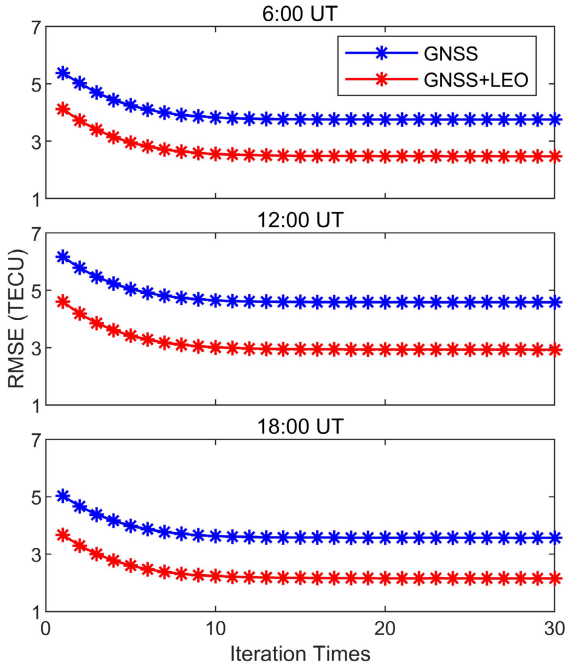


Fig. 5. Variations of RMSEs of CIT_G and CIT_{GL} with iteration times, at 6:00, 12:00, and 18:00 UT on DOY 308, 2021.

moments. As can be seen, the errors of CIT_{GL} results are distinctly smaller than those of CIT_G ones over the ocean areas (see red dashed ellipses in Fig. 6), indicating that incorporating the LEO data improves the CIT accuracy over these areas. Furthermore, above 800 km, where the contribution of the LEO data to the total data volume is significant (as seen in Fig. 4), visible reductions in IED errors are obtained by CIT_{GL}.

The maps of VTEC errors derived from CIT_{GL}/CIT_G results with reference to the IED true values at the three moments are shown in Fig. 7. Compared with CIT_G VTEC, CIT_{GL} VTEC values are generally closer to the IED true values and are of smaller errors. The errors of CIT_G-derived and CIT_{GL}-derived VTEC range from −4.72 to 0.63 TECU and from −3.40 to 0.50 TECU at 6:00 UT, from −5.76 to 0.60 TECU and from −4.04 to 0.47 TECU at 12:00 UT, from −6.00 to 0.44 TECU and from −4.27 to 0.39 TECU at 18:00 UT, respectively. Compared with CIT_G, the improvements of CIT_{GL} on the VTEC accuracy are mainly obtained around the equatorial ionospheric anomaly (EIA) region and in the ocean areas.

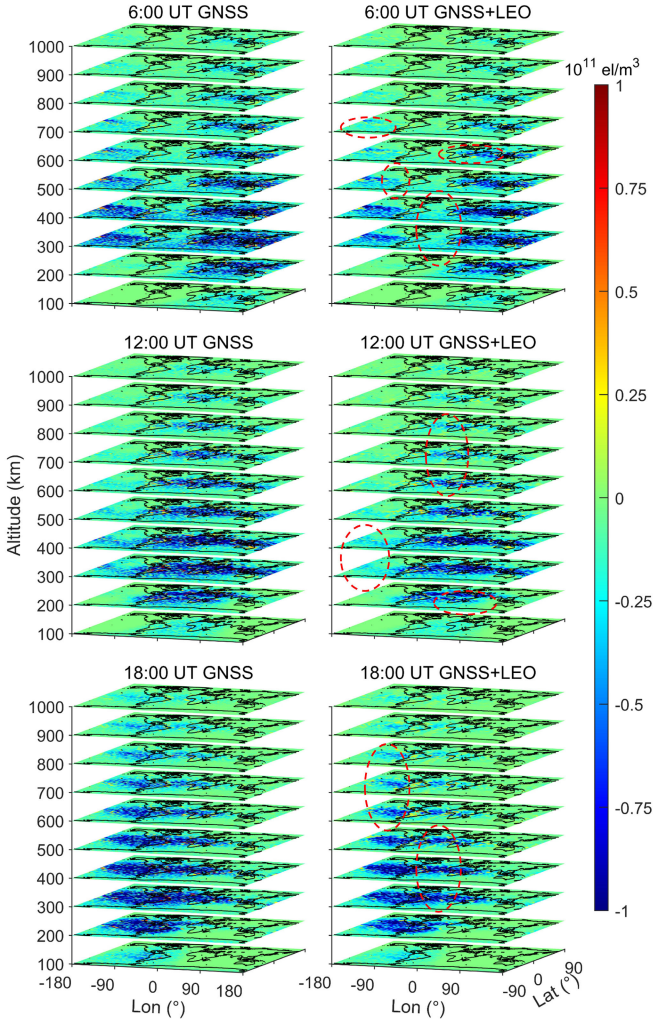


Fig. 6. Three-dimensional maps of the errors of CIT_{GL} and CIT_G results relative to the IED truth at 6:00, 12:00, and 18:00 UT on DOY 308, 2021.

TABLE II

RMSEs (Unit: TECU) of the STEC Derived From CIT_{GL} and Their Improvements Compared With CIT_G Results at 6:00, 12:00, and 18:00 UT on DOY 308, 2021

	CIT_{GL}	CIT_G	Improvement
6:00 UT	4.87	3.53	27.46%
12:00 UT	5.21	3.87	25.66%
18:00 UT	4.52	3.59	20.65%
Average	4.87	3.66	24.59%

B. Real Data Experiment

The real data experiments are conducted using both CIT_{GL} and CIT_G models during the geomagnetic storm occurring on DOY 308, 2021. The time series of Dst index in DOY 307–309, 2021 are presented in Fig. 8. Specifically, Dst remained above −20 nT in the quiet time on DOY 307. On DOY 308, Dst dropped sharply in the initial phase of the storm from 21:00 UT, DOY 307 to 8:00 UT, DOY 308, and further decreased from 10:00 to 13:00 UT, reaching the minimum of −105 nT and entered the main phase, afterward increased gradually in the recovery phase. Therefore, three moments, 6:00, 12:00, and 18:00 UT on DOY 308, 2021, are selected for the experiment, which respectively represent the initial, main, and recovery phases of the geomagnetic storm. The IED background is obtained from the IRI-plas model. The STEC derived from the reconstruction results of CIT_{GL} and CIT_G is also validated using the randomly selected reference STEC. The RMSEs of the STEC at the three moments are presented in Table II, which are respectively 3.53, 3.87, and 3.59 TECU for CIT_{GL} and 4.87, 5.21, and 4.52 TECU for CIT_G. The average improvement on the accuracy of STEC obtained by CIT_{GL} is 24.59%. The validation results show that the CIT_{GL} STEC is generally closer to the reference STEC than the CIT_G STEC, which means that the CIT_{GL} outputs agree with the real ionospheric state better.

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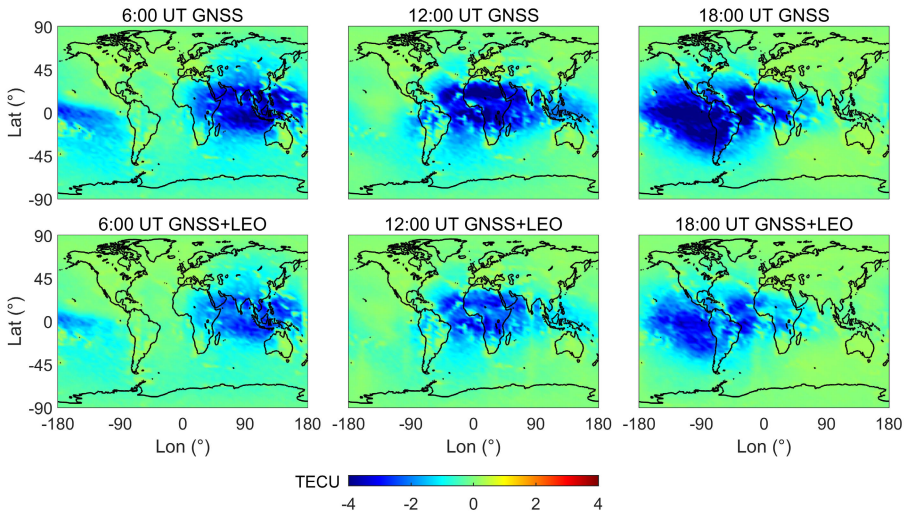


Fig. 7. Maps of VTEC errors derived from CIT_{GL}/CIT_G results at 6:00, 12:00, and 18:00 UT on DOY 308, 2021.

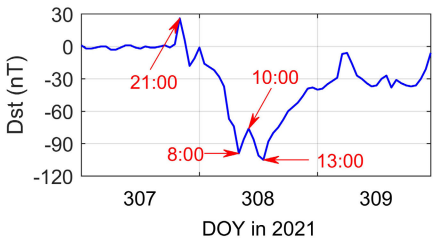


Fig. 8. Time series of the Dst index in DOY 307–309, 2021.

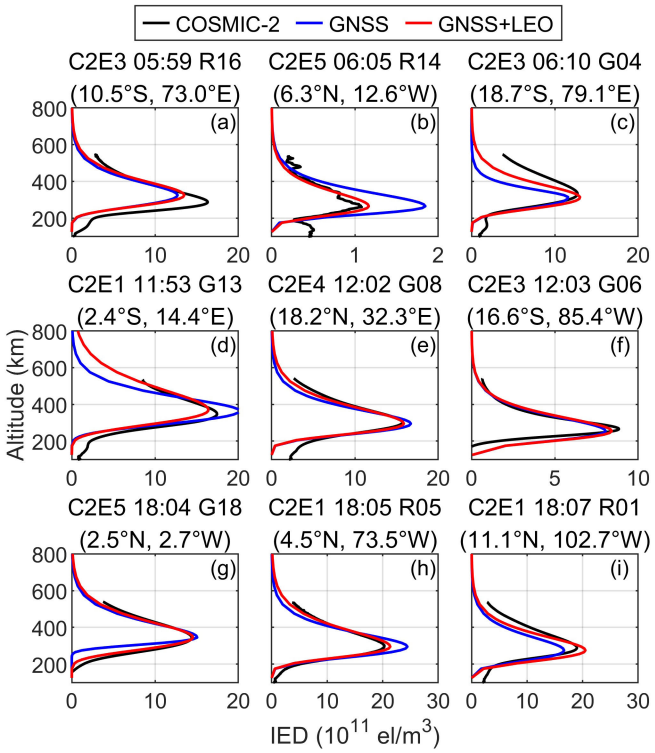


Fig. 9. Comparisons of the IED profiles from CIT_{GL}, CIT_G, and COSMIC-2 around (a)–(c) 6:00 UT, (d)–(f) 12:00 UT, and (g)–(i) 18:00 UT on DOY 308, 2021. The title of the upper axis of each subfigure denotes the ID of the LEO satellite, the time of the RO event, the ID of the occulted GNSS satellite, and the location of the RO IED profile.

Fig. 9 shows the comparisons of the IED profiles from CIT_{GL} and CIT_G results with those from COSMIC-2 at each of the three moments. Note that the COSMIC-2 IED profiles are from the "ionPrf" product of the COSMIC-2 mission, which are different from the STEC observations from the "podTEC" product and are not involved in the CIT experiment. It can be seen that for most of the cases, the IED profiles from the CIT_{GL} results agree better with the COSMIC-2 profiles than those from the CIT_G results. In Fig. 9(b)–(e), (h), and (i), the CIT_{GL} IED profiles demonstrate better agreement with COSMIC-2 ones than CIT_G IED profiles above hmF2, which is due to the higher voxel penetration rate above 400 km after adding LEO data. Besides, the improvements of CIT_{GL} IED profiles over CIT_G ones are generally greatest near hmF2, and the F2 peak electron density (NmF2) of CIT_{GL} is generally closer to COSMIC-2 observations than that of CIT_G, which are especially distinctly demonstrated in Fig. 9(b), (d), and (h). However, in some cases, the improvements of the CIT_{GL} IED profiles compared with CIT_G ones are not significant, such as shown in Fig. 9(a) and (f), which is due to that there is no LEO data or too little LEO data in these places.

Fig. 10 further presents the comparisons of NmF2 and hmF2 from CIT_{GL} and CIT_G results with those from COSMIC-2 observations within the CIT time window. For all of the three moments, the RMSEs of CIT_{GL}-derived NmF2 and hmF2 values are generally lower than those of CIT_G-derived ones, and the reduction in the RMSE of CIT_G-derived NmF2 at 6:00 UT is the most distinct, reaching 36.93%. Note that there are some outliers of CIT NmF2 and hmF2 in Fig. 10 that are quite different from those of COSMIC-2, which is possibly due to that the corresponding voxels of these outliers are not traversed by any rays, and thus the NmF2 and hmF2 values of these voxels are not corrected and remain the same as the background. On the other hand, the numbers of outliers in CIT_{GL} results are significantly less than those in CIT_G results, especially for NmF2, which is also attributed to the reduction of empty voxels by CIT_{GL}.

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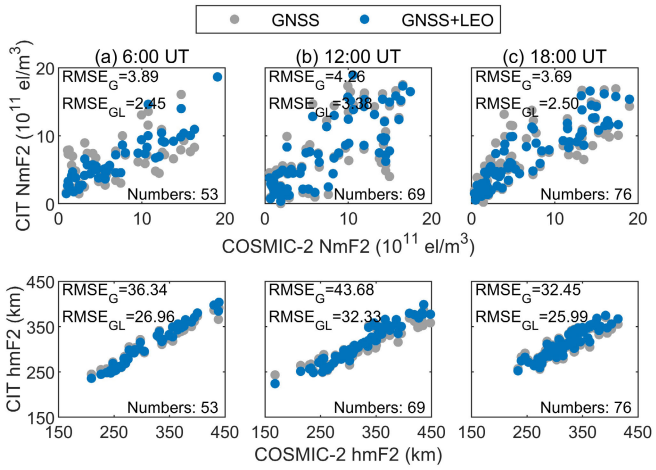


Fig. 10. Comparisons of NmF2 and hmF2 from CIT_{GL}, CIT_G, and COSMIC-2 around (a) 6:00 UT, (b) 12:00 UT, and (c) 18:00 UT on DOY 308, 2021. The number of observations, the RMSEs of NmF2 (unit: 10^{11} el/m^3) or hmF2 (unit: km) from CIT_G ($RMSE_G$) and CIT_{GL} ($RMSE_{GL}$) are shown in the upper left corner of each subfigure.

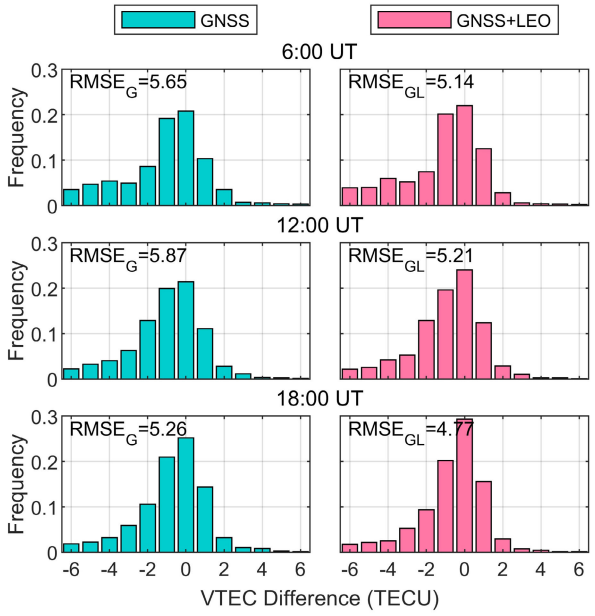


Fig. 11. Histograms of the difference between the global VTEC maps derived from CIT_{GL}/CIT_G and those from GIM at the three CIT moments. The RMSE (unit: TECU) of CIT_G VTEC ($RMSE_G$) or CIT_{GL} VREC ($RMSE_{GL}$) is shown in the upper left corner of each subfigure.

The global VTEC maps derived from CIT_{GL} and CIT_G results at the three moments are testified by the gridded VTEC from the global ionospheric map (GIM), and the histograms of the difference between them are shown in Fig. 11. Compared with the differences between CIT_G VTEC and GIM VTEC, those between CIT_{GL} VTEC and GIM VTEC have a slightly higher degree of aggregation centered on zero at all the three moments. The RMSEs of VTEC at the three moments are all slightly reduced by CIT_{GL} over CIT_G, decreasing from 5.65/5.87/5.26 TECU to 5.14/5.21/4.77 TECU.

Fig. 12 further shows the 3-D distributions of the differences between the CIT_{GL}-derived and CIT_G-derived IED at the three CIT moments. It can be seen that the larger IED differences are mainly distributed in the EIA zone and the ocean areas in the altitudes between 200 and 600 km. In other regions and altitudes, the IED differences are generally much smaller.

The results presented in Table II and Figs. 11 and 12 suggest that fusing GNSS and LEO data mainly helps improving the CIT accuracy in the areas where the LEO data are abundant (the EIA zone) or where the ground-based GNSS data are limited (the ocean areas), which correspondingly brings to the overall improvement of the CIT performance of the globe. On the other hand, as shown in Table II and Figs. 9–11, although the RMSEs of all the evaluated ionospheric parameters at 12:00 UT (in the main phase of the storm) are generally slightly higher than those at 6:00 UT and 18:00 UT, the improvements of CIT_{GL} over CIT_G at 12:00 UT have no distinct degradations compared with those at the other two moments. This indicates that even under severe storm conditions, CIT_{GL} is still effective in improving the CIT performance.

C. Analysis of the Geomagnetic Storm

In this section, CIT_{GL} model is used to analyze the ionospheric disturbances during the geomagnetic storm on DOY 308, which is derived by subtracting the TEC and IED between the same moments of DOY 308 (storm day) and DOY 307 (quiet day). Fig. 13 shows the global VTEC differences from 2:00 UT to 16:00 UT, with a 2-h interval, which are calculated based on the VTEC observations from the Madrigal Database and are used as the reference VTEC data. Compared with DOY 307, there are a considerable number of strong positive or negative ionospheric disturbances, called positive or negative storm effects, occurring over some specific regions on DOY 308. First, the VTEC over South America had significant decreases reaching 10 TECU from 2:00 to 6:00 UT, known as the negative storm effect. Second, VTEC increased over the European region at 6:00 UT, further increasing to higher than 10 TECU from 8:00 to 10:00 UT, and after 12:00 UT, the VTEC increasements were reduced to about 5 TECU. Third, two peaks were found in the strong VTEC increasements over East Asia between 110° and 150°E, respectively, occurring at around 6:00 and 10:00 UT, and the two positive ionospheric disturbances were also observed by Singh et al. [29]. Finally, over South America, positive ionospheric disturbances were presented by VTEC increasements since 12:00 UT. The magnitude and the spatial coverage of the VTEC enhancement further increased until 18:00 UT (not shown in Fig. 13). As pointed out by Shahzad et al. [30], the VTEC increasements in DOY 308 compared with the previous day, which occurred over both East Asia and South America, were parts of the enhanced EIA that moved westward with Earth's rotation during the storm time.

To preliminarily assess the capability of the CIT_{GL} model to monitor the ionospheric variations during the storm, the global VTEC differences derived from the CIT_{GL} results are shown in Fig. 14. Fig. 14 clearly illustrates the VTEC disturbances over the European region, which began to increase at 6:00 UT and gradually extended from south to north. This figure also

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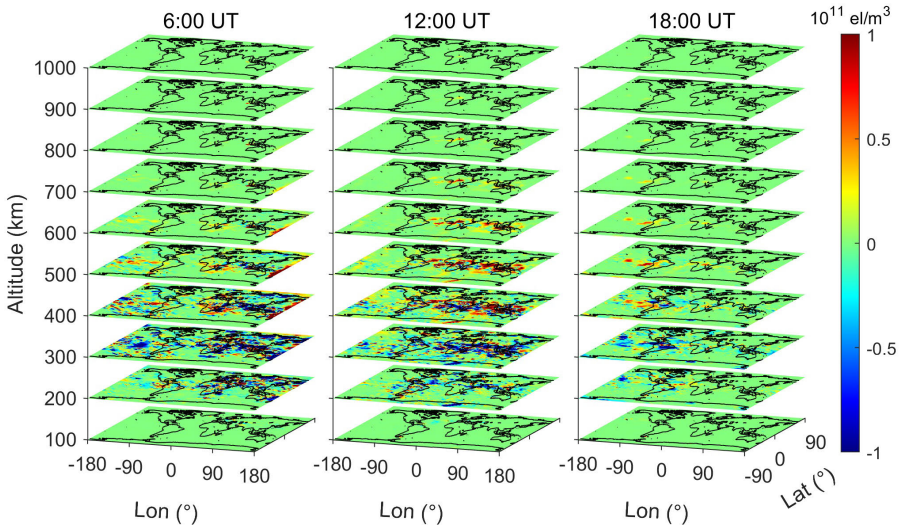


Fig. 12. Three-dimensional maps of the differences between CIT_{GL}-derived and CIT_G-derived IED at 6:00 UT, 12:00 UT, and 18:00 UT on DOY 308, 2021.

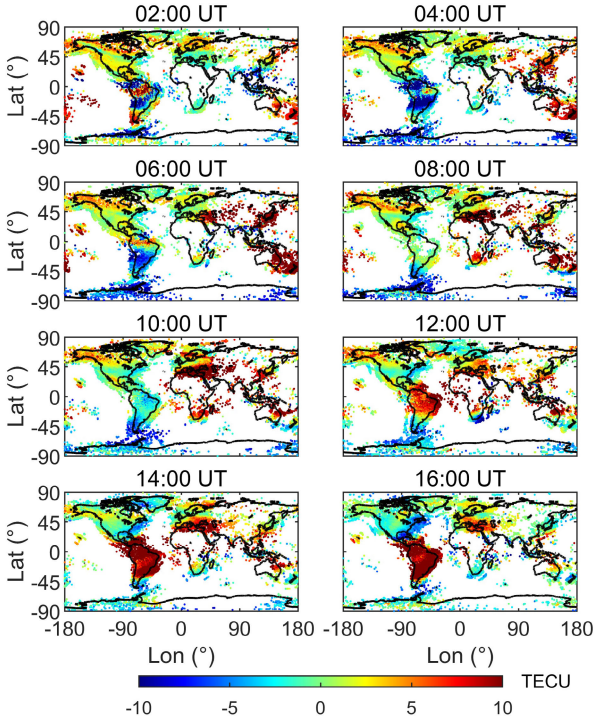


Fig. 13. Latitude–longitude maps of global VTEC differences at the same moments from 2:00 to 16:00 UT between DOY 308 (storm time) and DOY 307 (quiet time), with a 2-h interval, calculated based on the VTEC observations from the Madrigal Database.

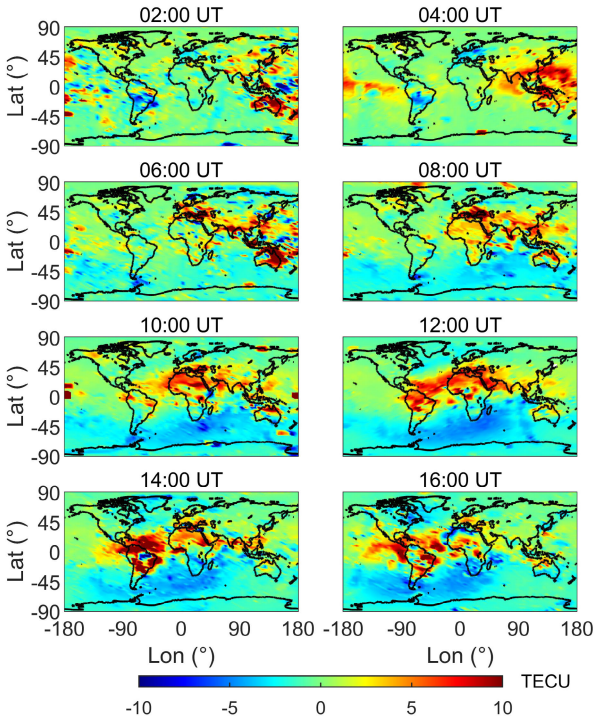


Fig. 14. Same as Fig. 13, but derived from CIT_{GL} results.

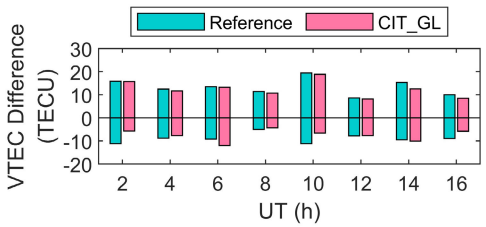


Fig. 15. Variation ranges of the VTEC differences at the same moments from 2:00 to 16:00 UT between DOY 308 (storm time) and DOY 307 (quiet time), with a 2-h interval, derived from the reference VTEC observations from the Madrigal Database and CIT_{GL} results, respectively.

shows the negative (positive) VTEC disturbances occurring over South America at around 2:00 (12:00) UT on DOY 308. In addition, the peaks of VTEC differences over East Asia at 6:00 UT are clearly reconstructed by CIT_{GL}, although the peaks observed at 10:00 UT are slightly smaller than those presented in Fig. 13. Furthermore, similar to Fig. 13, Fig. 14 also traces the westward process of the enhanced EIA. The agreements between the distributions of the ionospheric disturbances in Figs. 13 and 14 indicate the ability of the CIT_{GL} model to monitor both the positive and negative storm effects.

Since the magnitudes of the ionospheric disturbances can be quantitatively reflected by the variation ranges of the VTEC differences between the storm day and quiet day, Fig. 15

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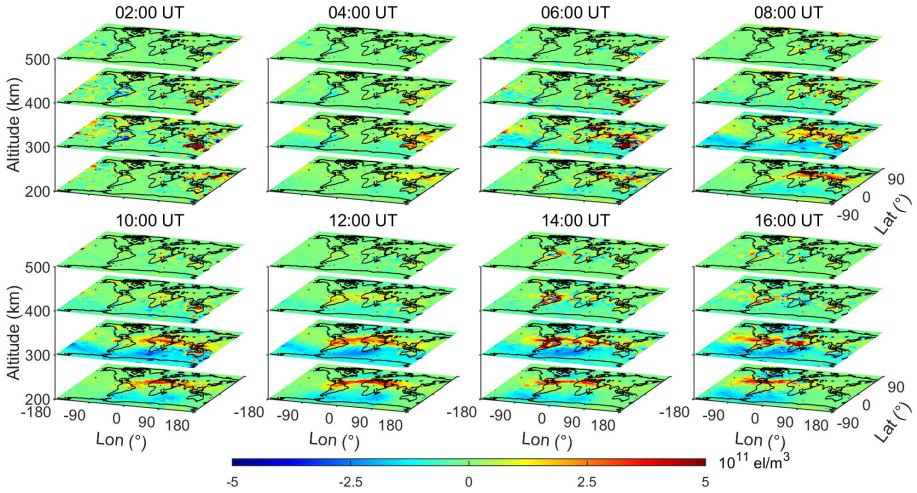


Fig. 16. Three-dimensional maps of the global IED differences at the altitudes from 200 to 500 km, at the same moments from 2:00 to 16:00 UT between DOY 308 (storm time) and DOY 307 (quiet time), with a 2-h interval, which are derived from CIT_{GL} results.

further presents the variation ranges of the VTEC differences derived from the reference VTEC and CIT_{GL} results, respectively. The variation ranges derived from CIT_{GL} results agree well with those derived from the reference VTEC, and the differences between the two types of variations ranges are generally less than 5 TECU. This quantitatively validates the accuracy of CIT_{GL} in monitoring the VTEC disturbances during the storm.

Compared with the VTEC maps that only show the horizontal distribution of the ionospheric disturbances, CIT has the capability of further revealing the more detailed view of the 3-D distribution of the localized IED disturbances caused by the geomagnetic storm. Fig. 16 shows the 3-D maps of global IED differences between the same moments of DOY 308 and DOY 307, with a 2-h interval, which are also obtained by CIT_{GL}. Here, we only present the 3-D maps of IED difference at the altitudes from 200 to 500 km, which is due to that during the storm, the ionospheric disturbances beyond this altitude range are much smaller. The IED differences were the most significant at the altitudes of 200–300 km, and it can be inferred that the VTEC disturbances over Europe and East Asia were mainly attributed to the IED enhancements at this altitude range. For the two peaks of TEC disturbances over East Asia, which were observed at 6:00 and 10:00 UT, the corresponding IED enhancements also appeared at the altitude of 400 km. In South America, at 2:00 UT, large negative IED disturbances occurred at the altitude of 300 km, while after 12:00 UT, large positive IED disturbances spread over the broad altitude range of 200–400 km.

As mentioned above, the enhanced EIA during the storm time moved westward with time. Before 10:00 UT, EIA was mainly distributed at 200–300 km, which was also demonstrated in the IED disturbances at these altitudes over Europe and East Asia. While after 12:00 UT, EIA gradually extended upward above 400 km. At 14:00 UT, over South America, the magnitudes of the IED disturbances at 400 km reached almost the same level as those at 300 km. As shown in Fig. 8, the main phase of the geomagnetic storm occurred around 13:00 UT, when Dst index reached the minimum of

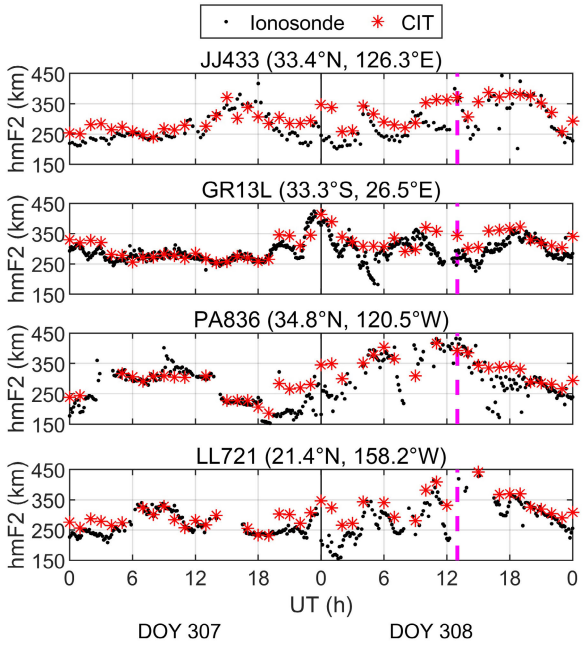


Fig. 17. hmF2 observations from four ionosonde stations in the EIA zone and those derived from CIT_{GL} on DOY 307 and DOY 308, 2021. In each subpanel, the name and the location of the station are denoted over the top horizontal axis, and the pink dashed line marks the time when Dst index reaches the minimum.

−105 nT. Theoretically, in the main phase of the storm, the ionospheric plasma and the hmF2 in the EIA zone should be uplifted relative to those in quiet time, and this phenomenon has been widely observed [31], [32], [33]. Fig. 17 further illustrates the hmF2 observations from four ionosondes in the EIA zone and those derived from CIT_{GL} on DOY 307 and DOY 308. For all the four ionosondes, the hmF2 observations at a specific time of DOY 308 were generally higher than those at the same time of DOY 307, which is most distinct around the main phase of the storm (pink dashed lines in Fig. 17). CIT_{GL} results captured the uplift of hmF2, which is in good agreement with the ionosonde observations, demonstrating the advantage of CIT in monitoring the vertical variations of the

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ionospheric disturbances. Note that the uplift of hmF2 during the main phase of this storm, which was dominated by the prompt penetration electric fields and may be attributed to the thermospheric composition variation, was also reported by Zhai et al. [34] by using ionosonde observations. Here, based on the outputs of CIT_{GL}, we further present the associated temporal evolution of IED disturbances over the globe, which should contribute to a thorough understanding of the influence of the storm on the ionosphere.

V. Conclusion

In this study, we present a global CIT model fusing ground-based GNSS and space-borne LEO satellite data. STEC data from more than 600 ground-based GNSS stations and six space-borne RO missions including a total of 34 LEO satellites are jointly used as the data source for CIT_{GL}, and the VTEC data are also combined with STEC data to further improve the data volume. Compared with the global CIT model using only ground-based GNSS data, namely CIT_G, CIT_{GL} increases the voxel penetration rate, especially in the ocean area, and thus improves the accuracy of the CIT results. The performance of CIT_{GL} is verified in both the simulation experiment and the real data experiment, with reference to independent STEC data, COSMIC-2 IED profiles, and GIM VTEC. Furthermore, CIT_{GL} is applied to monitor the ionospheric disturbances during the geomagnetic storm on DOY 308, 2021.

Specifically, the simulation experiment demonstrates that the voxel penetration rate in the CIT is improved by 18.31% via adding the LEO data, which fills in some voxels not traversed by ground-based GNSS rays. The CIT_{GL} results are much closer to the IED true values than CIT_G results, and CIT_{GL} has the higher convergence rate than CIT_G during the iteration process. In the real data experiment, the RMSEs of the STEC from CIT_{GL} results at the three moments are all reduced compared with those from CIT_G results, reaching an average reduction of 24.59%. Most of the IED profiles from CIT_{GL} results are more consistent with the COSMIC-2 IED profiles than those from CIT_G results, and the accuracies of the reconstructed NmF2 and hmF2 are also improved by CIT_{GL}. Besides, the global VTEC maps derived from CIT_{GL} results are closer to GIM VTEC than those derived from CIT_G results. Compared with CIT_G, CIT_{GL} mainly improves the CIT accuracies in the areas where the LEO data are abundant or where the ground-based GNSS data are limited, which correspondingly brings to the overall improvement of the CIT performance of the globe. Around the geomagnetic storm on DOY 308, 2021, the maps of global VTEC disturbances derived from CIT_{GL} are basically consistent with those from the reference VTEC observation data. CIT_{GL} effectively reconstructs the 3-D distribution and the evolution of the localized IED disturbances, demonstrating the westward process of EIA. In particular, CIT_{GL} also captures the uplift of the hmF2 during the main phase of the storm.

In summary, fusing the ground-based GNSS and the space-borne LEO satellite data helps to increase the voxel penetration rate in the CIT and brings improvement in the performance of the global CIT model. Currently, a considerable number of space-borne LEO satellites have been put in operation, which is a valuable complement to the ground-based GNSS. It is expected that in the future, as more LEO satellites are deployed, the CIT model can be further improved in precision and resolution, which should be a powerful tool for understanding ionospheric activity and space weather.

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